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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
M.Sc Degree Examination – APRIL 2021

PMT703 – ORDINARY DIFFERENTIAL EQUATIONS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Find the general solution of the equation $x'' - \frac{2}{t^2}x = 0$, $0 < t < \infty$.

2. Define Laplace transform of f(t).

3. Solve $x_1^1 = 5x_1 - 2x_2$
$x_2^1 = 2x_1 + x_2$

4. Define Non-homogeneous linear system.

5. Write the Lipschitz condition.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Solve : $x'' + x' + 2x = 0$, $x(0) = 0$, $x'(0) = 2$.

7. Solve : $x'' - 2tx' + 2x = 0$.

8. Prove that if $\Phi(t)$, $t \in I$ be a fundamental matrix of the system $x' = Ax$ such that $\Phi(0) = E$, where A is a constant matrix. Here E denotes the identity matrix then Φ satisfied $\Phi(t+s) = \Phi(t)\,\Phi(s)$ for all values of t and $s \in \mathbb{R}$.

9. Determine e^{tA} and a fundamental matrix for the system $x' = Ax$ where $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 5 \end{bmatrix}$.

10. Prove that $|x(t) - x_n(t)| \leq \frac{L(Kh)^{n+1}}{K(n+1)!} e^{Kh}$, $t \in [t_0, t_{0+h}]$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Solve the differential equation $x'' + b_1 x' + b_2 x = 0$ and verify that the solutions are linearly independant.

(or)

b) Find the solution of the equation $x'' - x = 1$, which vanishes when $t = 0$ and tends to a finite limit as $t \to \infty$.

12. a) Solve : $2x'' + x = 2t^2 + 3t + 1$, $x(0) = x'(0) = 0$.

(or)

b) Use Laplace transforms to solve the IVP $x'' - x' - 2x = 3e^t$, $x(0) = 1$, $x'(0) = 0$.

13. a) Prove that the set of all solutions of the system $x' = A(t)x$ on I forms on n-dimensional vector space over the field of complex numbers.

(or)

b) Find the determinant of the fundamental matrix ϕ which satisfies $\phi(0) = E$ for the system $x' = Ax$ where $A = \begin{bmatrix} 1 & 3 & 8 \\ -2 & 2 & 1 \\ -3 & 0 & 5 \end{bmatrix}$.

14. a) Let Φ be a fundamental matrix for the system $x' = A(t)x$ for $t \in I$ then prove that $\Phi(t) = \Phi(t) \int_{t_0}^{t} \Phi^{-1}(s)b(s)ds$, $t \in I$ is a solution of the IVP $x' = A(t)x + b(t)$, $x(t_0) = 0$.

(or)

b) Determine e^{tA} and a fundamental matrix for the system $x' = Ax$ where $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & -2 & 1 \\ 0 & 3 & 0 \end{bmatrix}$.

15. a) State and prove Picard's theorem.

(or)

b) Prove that the IVP $x' = f(t, x)$, $x(t_0) = x_0$ has a unique solution defined on $t_0 \leq t < t_0 + h$, $h > 0$ if the function $f(t, x)$ is continuous in the strip $t_0 \leq t \leq t_0 + h$, $|x| < \infty$ and satisfied the Lipschitz condition $| f(t, x_1) - f(t, x_2) | \leq K |x_1 - x_2|$, $K > 0$, K being the Lipschitz constant.

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